School of Engineering and Materials Science

Handbook for MSc Extended Research Projects

90 Credit Project Module EMS715P

2024-2025

Module Organiser:
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1. Overview & Aims of the MSc Extended Research Project Module

This project module gives you an opportunity to carry out an in-depth extended research project that spans all three semesters of your MSc programme. Undertaking an **extended research project** is a significant component of your MSc degree and requires you to independently conduct research of significant scope and depth whilst under academic supervision.

Your **MSc Extended Research Project** will address specific problems and/or challenges within a particular area of interest in science and engineering. You will have an opportunity to build on your existing scientific and engineering knowledge; and to develop a range of other research project skills and abilities e.g., reviewing literature; critical analysis; project management; communication and organization skills; devising a research methodology; data analysis; academic writing; constructing logical arguments to justify pursuing a particular area of research, methodology and methods; and project planning to meet key deadlines.

MSc extended research projects are quite diverse across different domains of science and engineering. As such, extended research projects sit along a continuum of science and engineering research to **encompass both traditional research-based projects and industry-based projects.** However, extended research projects will still share many overlapping components and characteristics, whilst also accommodating distinctive elements suited to research and/or industry.

1.1 The key aims of conducting an extended research project at MSc level include:

1. To develop research skills e.g., critical analysis of literature; scientific writing and referencing styles; proposing rationales and defining research question(s); devising project methodology and executing a research project.
2. To develop students' understanding of research methodologies and the scientific method; and applying suitable methodology design to an extended research project.
3. To appreciate the ethical implications for conducting research; the environmental and societal impact of solutions to complex problems; and the need to minimize adverse impact of conducting research and research outcomes.
4. To develop transferable skills e.g., project management; project implementation and completion; working autonomously; collaborative working with research supervisor; and communicating research findings to a target audience.
5. To develop key competences and commitments relevant to professionalism in science and engineering and/or to work toward professional registration e.g., CEng.

1.2 The key learning objectives of the extended research project are to:

a. Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems in an innovative and original way.
b. Demonstrate independent thinking and working; initiative; and competent technical judgement in science and engineering.
c. Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles.
d. Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.
e. Select and evaluate technical literature and other sources of information to address complex problems.

f. Design solutions for complex problems that meet a combination of societal, user, business and customer needs as appropriate.
g. Apply the need to minimise adverse impacts of conducting research e.g., health & safety, diversity, inclusion, cultural, societal, environmental, commercial matters, codes of practice, and industry standards.
h. Evaluate the environmental and societal impact of conducting an extended research project; and the impact of outcomes or solutions to complex problems.
i. Communicate effectively on complex engineering matters with technical and non-technical audiences.

1.3 The final project report:

Your research project will culminate in the writing up of a final project report. The path to a final project report submission is likely to involve interacting with other people e.g., your project supervisor, technical staff, postgraduate students, academic staff, postdoctoral fellows and administrative staff. Below are examples of some key points to help progress you through your extended research project:

- Apply and develop your time management and planning skills.
- Critically review literature of previous work conducted by others.
- Plan for writing up your final project report e.g., create a report template; create key section headings.
- Use tables, figures, diagrams, charts and graphs to help communicate sections of your final report. Remember to signpost the reader to tables, figures, diagrams, charts and graphs in the body of your text.
- Generate results, tables, charts and graphs to help communicate.
- Incorporate appropriate calculations, simulations modelling.
- Present your results, findings and outcomes and be able to discuss their implications.
- Cite references accurately and consistently using the appropriate referencing convention and style.
- Present arguments and conclusions in an accessible, logical and structured manner suitable for both technical and non-technical peers.

The above are skills will also be sought by a potential future employer, and the extended research project provides to wit an excellent opportunity to develop these skills and add to your portfolio of achievements.

Note: *In order to graduate with an MSc you need to pass all taught modules and the Extended Research Project. Failure in modules to the value of not more than 30 credits (2 modules other than the Project) can be condoned if your average mark over all the taught modules is not less than 50%, and the mark for the two failed modules is not less than 40%. You will have the opportunity to resit your failed examinations at the end of the summer (Late Summer Resits).*
Please refer to the latest QMUL academic regulations for further details.

2. Assessment and Deadlines

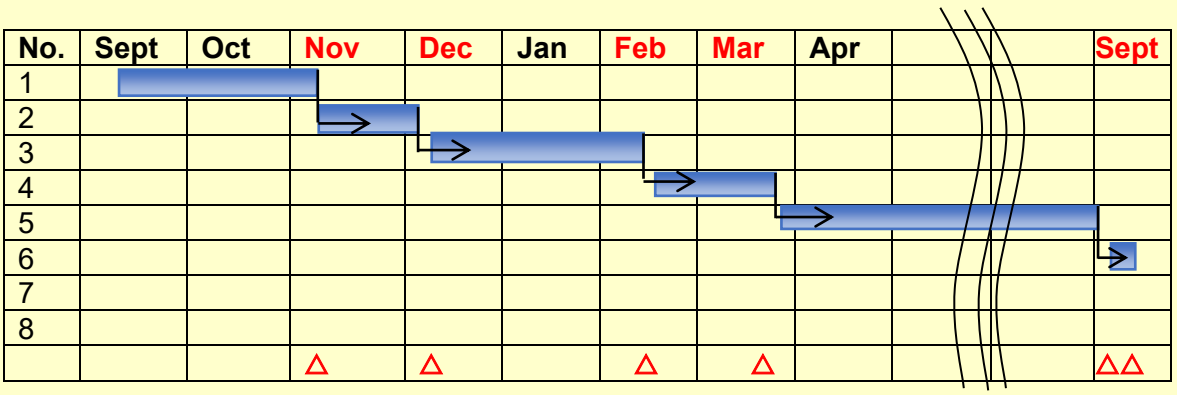
Your performance during the project is assessed by various means as outlined in the table below. Each component of assessment has its own deadline and all submissions are electronically submitted. If you do not submit an assessment by the deadline specified on QMPlus, your submission will incur a penalty and daily deduction in marks up to 7 days after which the assignment will receive ZERO marks. Please ensure that your project planning and contingency planning gives you enough time address any last-minute technical problems or mishaps to avoid submitting late and incurring a penalty. Please note that reasons for late submissions, such as, "*hard drive crashed*"; "*forgot to check emails*"; or "*QMPlus was running slowly whilst uploading*" cannot be accepted as valid reasons for late submissions.

Assessment Breakdown

	Component of Assessment	**Weighting**	***Submission Timeline**	**Assessed by-**
1	Directed Learning Assignment 1 (DLA1)	10%	Mid-November	Supervisor
2	Directed Learning Assignment 2 (DLA2)	10%	Beginning of January	MO/TFs
3	Project Rationale, Aims and Objectives (RAO) e-Poster (A1 Size)	10%	Mid-Feb	Supervisor
4	State of the Art report (SOTA)	15%	End-March	Supervisor
5	Final Report	40%	Beginning Sept	Supervisor and 2nd Examiner
6	Viva Examination	15%	Beginning Sept	Supervisor and 2nd Examiner

See also Appendix 3 & Appendix 4.

Submission Timeline



***△See QMPlus for exact deadlines dates.**

3. **Lectures, Workshops, Supervision & Self-Study**
   It is important to attend lectures and workshops as these sessions intend to help inform the direction you take for your MSc extended research project and your project plan. In addition, you will need to work independently on researching articles/readings, resources planning, project management and it is recommended that you arrange regular meetings with your supervisor. *Refer to the module QMPlus page*.

4. **Getting Started**
   You will be given an opportunity to select your own key areas of interest for your final year project and we will do our best to find an appropriate academic as your supervisor. We will also do our best to match your area of supervisor. We may not be able to guarantee a particular academic for your project because there is a limited number of academics to act as supervisors. However, in SEMS there are academics that have interdisciplinary knowledge and skills that can be adapted to different project interests.

   Note: *You cannot change supervisor without re-allocation by the module organiser. Project supervisor allocation depends on the availability and workload of academics*.

5. Project Log & Project Planning

A project log is a detailed record of all the work you are doing in your project where you bring your notes, ideas, planning and progress together in a single document. The logbook may also include concepts, sketches, equation developments, calculations, graphs, summaries of papers you have read including their full source details and records of meetings with your supervisor or others from whom you seek advice. The logbook (*aka* notebook) can be requested for review at any point during the year by your supervisor or academic assessors. You are strongly encouraged to maintain your logbook to help you, and your supervisor, make progress leading to a successful outcome from your extended research project. You may also be asked to upload a pdf of your log at certain points during the year.

The purpose of a project log is to:

- Maintain a summary record of the work you have undertaken on at least a weekly basis;
- Develop structuring and planning skills (project timeline);
- Identify directions for research/work;
- Maintain a record of progress/milestones achieved;
- Identify problems (supervisor or student) in a timely fashion.

The project log includes the following (although this list is not exhaustive)

- Project Title
- Project Aim(s)
- Project Objectives
- Project Plan
- Record of supervision meetings
- Record of notes, discussions, thoughts, suggestions, ideas, questions/queries
- Record of progress and actions
- Record of resources, literature, reading, article etc.
- Record of sketches drawings, graphs, illustrations etc.

More guidance will be provided during module lectures, module sessions and supervision. Lectures/workshops will cover logbook formats, layout, content and ongoing logbook entries and regular logbook reviews/submissions.

Success on the MSc degree programme rests upon careful planning. Project planning and project management is a transferable skill which graduates are expected to possess. Balancing the work-load over the year involves placing emphasis on project work at the beginning of the year, leaving time for revision toward the end. It is important that you meet regularly with your supervisor and discuss your progress and plans for further work. On average, you should typically spend between 12 to 15 hours per week on your project and keeping your logbook up to date. You will be asked to update and submit your logbook during the year.

6. Role of the Project Supervisor

The role of your project supervisor is to support you whilst undertaking your MSc Extended Research Project. Your project supervisor will provide you with guidance and feedback on the direction and progress of your project as well as feedback on project coursework submissions. You can approach your supervisor to share and explore your project ideas and project progress, however, it is not the role of the supervisor to prescribe each stage of your project. Use the time during supervisor meetings effectively through preparation and planning (*see earlier section on logbooks & project planning*). For your MSc extended research project, you will be expected to take ownership of your project and adopt a proactive approach to arranging regular meetings with your supervisor and with your project planning. If you have problems contacting or meeting your supervisor, email the module organiser.

7. Technical Project Support

If you need models, fixtures or apparatus to be manufactured, it is essential that your request is made at the earliest possible date. Go to the following link for information on the workshop, laboratories and facilities in SEMS.
https://www.sems.qmul.ac.uk/research/facilities/mechanicalworkshop/whatwedo/
https://www.sems.qmul.ac.uk/pgadmissions/facilities/

See also the points below to help clearly communicate with technical support staff.

1. Obtain approval from your supervisor for the purchase of consumables (https://admin.sems.qmul.ac.uk/forms/technical/) via the School's Intranet.
2. For work requests, the project manufacturing requirements must be clearly defined and approved by your supervisor before you start to discuss any work with a technician. Only students who have an accepted and issued work request https://admin.sems.qmul.ac.uk/forms/technical/ to a technician will be able to

arrange a meeting time with the technician. This will enable the technician to prepare for any discussions and for the Laboratory and Workshop Manager to progress the work. The Laboratory and Workshop Manager will then inform you by email when to discuss the project with an appropriate technician.

3. Ensure that you plan ahead and provide the technical project support staff with clearly explained and structured requirements. For example, engineering drawings with a parts list/BoM should be generated preferably using CAD software to show with 2D projections and exploded/assembled 3D drawings.
4. Reasonable attention must be given to the target dates for completion of the work.
5. Work will be given priority on a first-come first-served basis.
6. Close contact with the allocated technician should be maintained and, if necessary, discuss and agree any design modifications/amendments/refinements. Design modifications cannot be made endlessly and major re-designs will be considered as new work requests.
7. Student access to the student workshop will be available as appropriate, depending on levels of student competence as determined by the Laboratory and Workshop Manager.
8. Record all discussions with your supervisor and the technicians in your project log.

8. Ordering Materials & Consumables

Ordering Consumables for Projects Consumables for all projects must be ordered by using the online https://admin.sems.qmul.ac.uk/forms/consumables/. Please complete the form making sure **all** appropriate areas have been filled in. When you have finished all sections click the 'Send this request' button and it will go to the designated supervisor for approval. Once the supervisor has approved the order it will go to SEMS Finance to be ordered.

Please note consumables should not be purchased in any other way.
Note. The University has preferred suppliers, which must be used. Requests for items that can be exempted from VAT must be directed to the Accounts Assistant. [In the past some students bought items themselves and then confronted the matter of VAT reclaim].

If, in exceptional cases, a new supplier is to be used, the name of the company must be added to the financial information system, Agresso, by Finance Department staff. Until the company's details are added to Agresso, the order cannot be placed. It is the duty of the order requester to obtain a sample of the company's headed paper [by fax is sufficient] so that the Accounts Assistant can complete the New Supplier's form for the Finance Department and attach the sample. Students should be mindful of the lead-in time in these circumstances.

MSc Extended Research projects have a budget of £50 allocated to them.

Note. Students should not make any purchases of consumables or equipment with their own financial resources in expectation of being reimbursed by the University, as this will not be authorised.

9. Project Module Assessments

9.1 Directed Learning Assignment 1 (DLA1)

The Directed Learning Assignment 1 (DLA1) provides you with a valuable opportunity to develop your research and analytical skills through extracting and evaluating articles of interest. For the DLA1, you will be asked to identify and carry out a concise critique of several research articles related to area of extended research interest. DLA1 enables you to critically analyses the rationale, content, outcomes and implications of each article through a process of evaluation and extraction. Guidance on the approach, structure and template will be given in advance of beginning DLA1.

9.2 Directed Learning Assignment 2 (DLA2)

The Directed Learning Assignment 2 (DLA2) develops your knowledge, awareness and understanding of ethical issues relating to research e.g., ethical principles, plagiarism, academic integrity, consequences of misconduct. DLA2 will assess your understanding through an online test.

9.3 Rationale, Aims and Objectives (RAO) A1 Size e-Poster

A clear and focused project rationale, with concise aims and objectives, is vital to gaining valuable data and insights from your extended research project. The justification behind the project and the design of your project methodology needs to also be subjected to critical analysis in order to review and evaluate the validity and reliability of your extended research project.

The RAO poster provides you with an excellent opportunity to showcase and communicate your project to both technical and non-technical audiences; and draw attention to the purpose, value, gap-in-knowledge and implications of your extended research project. Although the poster will be produced before the completion of your project, the design, layout and content of your poster will also help you to address the design of your project methodology and anticipate the type or kind of data and/or outcomes you will collect and analyse. The poster will also give you an opportunity to focus on the potential implications of project findings and conclusions in relation to the project rationale and its chosen methodology. Although the poster will be a 'snapshot' of your project, it is a valuable exercise to aid your project planning and clarify the purpose of your extended research project in a concise manner.

The Industrial Engagement Committee will also select suitable posters to showcase at the Industrial Liaison Forum (ILF) event held by the School of Engineering and Materials Science (SEMS) in the Octagon. The ILF event is an exciting and vibrant opportunity for our industrial partners and to view some of the best projects from this academic year, and to learn more about collaborating on projects with the School of Engineering and Materials Science (SEMS).

9.4 State of the Art Report (SOTA)

The State of the Art (SOTA) report critically reviews the literature on research, technology, methodologies, findings and conclusion research to allow the reader to understand how your SOTA informs the rationale behind the project. A literature review also places your extended research project within the context of existing knowledge and information from publications. The SOTA provides you with a valuable opportunity to build on your knowledge and critically review the research and technologies e.g., discuss and compare previous work and research designs as well as identify gaps in the current knowledge relevant to the focus of your extended research project.

The report should be written in a coherent and logical manner with arguments justified and supported by references for literature within the body of the report. The SOTA report should include:

- **Cover sheet.**
- **Declaration.**
- **Contents page.**
- **Brief introduction.**
- **Literature review.**
- **Conclusions and implications for SOTA for project RAO.**
- **List of references.**

The **maximum** length of the SOTA is **2000** words; and for the SOTA you can combine the cover sheet and declaration into a single page.

The SOTA report also gives you an opportunity to develop and improve your academic writing skills, as well as familiarise yourself with the appropriate format of a report and the referencing style. For example, 1.5 line spacing, 2.5 cm margins, headings and sub-headings,12pt Arial or New Times New Roman font, Harvard referencing. It is important that the formatting of reports is kept consistent throughout the report and complies with an appropriate referencing style and standard.

It is also recommended that you discuss and consult with your project supervisor about the appropriate report format and referencing style appropriate for your area of academic research. There are different referencing formats in academic writing, such as Harvard, Vancouver, MLA, Chicago, IEEE, APA, so it is important that you agree the format and style with your project supervisor and maintain consistency through your report writing.
Below are some links to help support you with your academic writing, but there are also many other sources accessible online.

Academic Skills Centre
https://www.qmul.ac.uk/library/academic-skills/
Academic Writing
https://aeo.sllf.qmul.ac.uk/topics/academic-writing/
Academic Style
https://aeo.sllf.qmul.ac.uk/writing/academic-style/
Referencing
https://aeo.sllf.qmul.ac.uk/writing/referencing/

9.5 Final Project Report

In addition to the points discussed in the previous section, guidance and details of the length, layout and format of the **final project report** are given in **Appendix 1** with the assessment criteria for the final project report in **Appendix 4**.

9.6 Viva Examination (Oral Examination)

Your viva (or oral examination) is an opportunity for you to discuss, explain and justify the work you did for your extended research project. The time and venue for your viva will be arranged between you, your MSc project supervisor and an independent examiner **after** submission of your final project report. Your viva will take place within the allowable period for conducting vivas, and not attending your viva has the same consequences as an absence from any other examination, i.e., the award of ZERO MARKS for this component. Only in exceptional circumstances where your absence is justified by acceptable independent evidence will a substitute time be arranged. The usual format is for your supervisor and the independent examiner to be present, and they will both have received your extended research report.

In advance of your viva, you will need to prepare a concise 3-5minute presentation of your extended research project – do not exceed a **maximum** of 5 minutes. Preparing a short presentation will also help you prepare for your viva by covering the salient points from your project report for your viva examiners. Your viva presentation could include, for example, a concise summary of your allocated project rationale, aims, objectives, specific problem(s), method/approach, justification, outcomes, results, findings and outcomes. Your summary of your extended research project needs to be short, clear and concise e.g., what you did; how and why you did it; what was the purpose and outcome (context, results & conclusion); why it was useful; and how did you address/meet the project RAO. Remember to upload a pdf file of your viva presentation to both viva submission points **before** your viva.

10. Queen Mary Logo

The Queen Mary logo is copyrighted and **MUST NOT** be reproduced by students. The presence of logo on the cover page or elsewhere in your report would indicate that the contents of these pages were produced and verified by Queen Mary University of London, which would not be the case

11. Academic Misconduct, Plagiarism and Academic Integrity

Academic misconduct is a serious examination offence with serious penalties applied for academic misconduct. Plagiarism is defined as the copying of material, text of figures, from another source without attribution. All photocopied, scanned or copied figures or text must have the source acknowledged.

The literature review and final report must be expressed in the student's own words and contain a significant individual contribution with acknowledgement and referencing of other literature or sources of information. Plagiarism involves presenting someone else's work as one's own irrespective of intention; close paraphrasing; copying work of another person or student; using the ideas of another person without proper acknowledgement; and repeating work that you have previously submitted without properly referencing yourself i.e., 'self-plagiarism' shall also constitute plagiarism.

QMUL will take all necessary steps to maintain the academic integrity of its programmes of study, and every allegation of academic misconduct will be dealt with in accordance with this policy. Plagiarism in reports is a serious examination offence and may incur

severe penalties e.g., receiving a zero mark; not graduating; suspension/expulsion and wasting an academic year.

Academic misconduct does not just involve plagiarism and includes the following:

- Examination offences.
- The use, or attempted use, of ghost-writing services for any part of assessment.
- The submission of work, or sections of work, for assessment in more than one module or assignment (including work previously submitted for assessment at another institution).
- The fraudulent reporting of source material, experimental results, research, or other investigative work.
- Collusion in the preparation or production of submitted work.
- Use, or attempted use, of ghost-writing services for any part of assessment – please note that having a third party proofread and change grammar and spelling may fall into this category as well.

For further information go to:
https://www.qmul.ac.uk/governance-and-legal-services/student-appeals/academic-misconduct/

12. Safety Considerations

All projects must be conducted and all project students must work in accordance with the current School Safety Regulations. For students whose project will involve experimental work, a **risk assessment for the project** must be drawn up and authorized by the supervisor. If the work involves working with chemicals a **COSHH form** must also be completed. **The student will not be allowed to embark on any experimental work until all necessary forms have been completed**.

See also http://www.sems.qmul.ac.uk/healthandsafety/ and **Appendix 3 – Laboratory Safety Regulations.**

Remember that safety is your responsibility and that it is extremely important to follow the regulations. However hard you are working, food and drink must **not** be brought into laboratories.

APPENDIX 1

Guidance for the Final Project Report Structure

A1.1 Length of Project
The final report should **not exceed** 6,000 words, but this word count **excludes**:

- **Cover sheet**
- **Abstract**
- **Declaration**
- **Contents and list of symbols**
- **Acknowledgements**
- **List of References**
- **Tables and Figures**
- **Appendices**

You will be asked to submit and electronic version of your final project report in a pdf format. Submission is made via QMPlus and you will need to submit a copy to two submission points, i.e., for the project supervisor and the independent examiner.

Note:** ***Units and Mathematical Equations
SI (Système International) units should be adopted in the report. Units of other systems, if used, should be placed within parentheses after the equivalent SI units. Symbols of a mathematical equation should be adequately defined. Care should be given in writing superscripts and subscripts.

A1.2 Sections of the Final Project Report

Your final report may be written in several different styles. This should be decided on in consultation with your project supervisor. Below is some guidance on the

- Title Page
- Declaration
- Abstract

- List of Contents
- List of Symbols
- Project Brief

- Introduction
- Literature review

- Main body of work (probably several sections, with appropriate titles to cover, say, rationale, aims, objectives, methodology and procedure)

- Results
- Analysis & Findings
- Conclusions
- Future Work

- References
- *Acknowledgements*
- *Other contributors to project*
- *Appendix*

***Research Paper Format:** *In some cases, and depending on your project supervisor, you may adopt a research paper format that complies with a particular journal paper's layout, structure, referencing and formatting. For example, technical journal papers are prepared* along the guidelines for IEEE journal articles.

A1.3 Front Cover

The front sheet should contain only the title of the project, the name of the author, and the month and year and name of the project supervisor. For example:

Queen Mary University of London
School of Engineering and Materials Science

Magnetically Aligned Graphite Particles in Electrodes for Fast Charging Lithium-Ion Batteries

By Anon Student
Student ID: 12345678

MSc Research Project (Xxxxxxx Engineering)

XXth August 2024

Supervisor: Dr Anon Academic

A1.4 Declaration

The declaration should immediately follow the title page and follow the style given below.

School of Engineering and Materials Science

MSc in Xxxxxxxx Engineering
EMS715P Extended Research Project
AUGUST 2024

DECLARATION

This report entitled

Magnetically Aligned Graphite Particles in Electrodes for Fast Charging Lithium-Ion Batteries

Was composed by me and is based on my own work. Where the work of the others has been used, it is fully acknowledged in the text and in captions to table illustrations. This report has not been submitted for any other qualification.
Name: Anon Student

Signed .....

Date .....

APPENDIX 2

Queen Mary
University of London

LABORATORY SAFETY REGULATIONS

There are particular hazards and risks associated with work in the School of Engineering and Materials Science.

The following regulations must be observed by all students using the facilities of the SEMS for their own safety and for the safety of others. These regulations are in addition to the general College Safety Regulations. If you do not observe the regulations you may be penalised by the College, but you will also be breaking the law and you could be fined or even imprisoned.

You will not be allowed to work in SEMS Laboratories until you have signed and returned a declaration. You should have already completed this in your 1st year of entry to the School.

GENERAL

1. It is the duty of each individual working in the School, in the College or on a Departmental Field course to behave responsibly and to take reasonable care for the safety of themselves and of other persons who may be affected by their acts or omissions.

2. Undergraduate students are not permitted to work in any room or laboratory at any time outside scheduled classes unless the responsible technician or academic staff member is present.

3. Postgraduate students who wish to work outside the normal working hours of 8.00-6.30 must comply with the current School directives on out-of-hours working (See SEMS Health and Safety Handbook).

4. Nobody is permitted to work alone where any potentially dangerous situation exists, such as electrical hazard, explosion or implosion hazard, chemical hazard, or mechanical or structural hazard. Exceptionally, working alone in laboratories may be authorised by written permission only by a supervisor or head of School after appropriate risk assessment. As a guide, hazards are deemed to exist in any situation not normally found in domestic households.

5. The School Safety Advisor must be informed in advance of any proposed work which may involve unusual or severe hazard whether in or outside the college. Risk assessments must be carried out for all laboratory or workshop procedures.

6. It is the duty of every individual to know the procedure for calling emergency services (3333 from any internal phone,9-999 if the College exchange fails), and to be familiar with the College instructions “What to do in an Emergency” and “Fire Precautions” displayed in the main entrances of all buildings. If the Fire Alarm sirens sound evacuate the Engineering Building via the nearest Fire Exit, follow instructions given by fire

marshals and proceed to the Assembly Point in Geography Square. Do not re-enter the building until a Fire brigade or College Security Officer has given the all clear to do so.

8. First aid boxes, fire extinguishers, fire alarms and telephones are installed throughout the School. Everybody should be familiar with the installations nearest to their work locations.
9. Nobody shall carelessly or intentionally interfere with or misuse anything that is provided in the interests of health, safety or welfare.

10. Any accident involving personal injury, which requires medical attention, must be reported to the Medical Centre and the School Safety Officer. Staff members must also know the procedure for reporting dangerous occurrences and serious personal injury accidents.

WORKING IN LABORATORIES

1. You should provide yourself with protective clothing, such as overalls or a laboratory coat, and to wear them at all times when working in the laboratories. The School cannot take any responsibility for soiled or damaged clothes resulting from not wearing protective clothing.

   You must wear stout, sturdy boots or shoes; **shoes made from textile material, open heal or open toes, high heal shoes and sandals are not allowed**.

   You will be asked to leave the laboratory if you do not wear suitable foot wear.

2. Chemicals are particularly hazardous. You should not handle or use any chemicals until you have been given the proper safety precautions and read the appropriate SDS (Safety Data Sheets) and completed a risk assessment. The following precautions must always be observed when handling chemicals:

   2.1 Protective clothing, gloves and goggles must be worn.

   2.2 No parts of the body should be touched whilst wearing gloves.

   2.3 Gloves should be washed before removal.

   2.4 Hands should be washed before eating, drinking or touching other parts of the body.

3. No machine, engine or equipment is to be used until you have been fully instructed in its use, as incorrect operation may be dangerous. If in doubt ask a member of staff.

4. You are only allowed in the laboratories during scheduled classes, unless prior permission has been obtained from a member of staff of SEMS. Project students must arrange laboratory working times with the Laboratory and Workshop Manager or the technician in charge of that laboratory.

5. Under no circumstances may anyone use equipment in a laboratory unless at least one other person is present. This is to ensure that aid will be readily available in the event of an accident.

6. Smoking, eating and drinking are not permitted in any laboratory area.

7. Orderly behaviour is essential, e.g. no running and messing around in laboratories.

8. All injuries, even minor ones such as cut fingers, must be reported to the Faculty Safety Advisor and treated by a qualified Firstaider (dial 3333 on any internal phone and ask for a Firstaider to be contacted).

9. Lifting must not be carried out if injury is likely to result; if in doubt ask for advice.

10. Care must be taken when using compressed air: IT CAN KILL.

11. Fume cupboards and fume extractors must be used to clear fumes from work areas.

12. Ear protection (i.e. ear plugs or ear defenders) must be worn in areas of high noise (>85 dbA).

13. Students with long hair must have it tied back.

14. Loose clothing (e.g. ties) and jewellery (e.g. watches, rings, necklaces, armbands) should not be worn in areas where rotating machinery is used. Laboratory coats should be buttoned up.

15. Safety glasses or goggles must be worn for all laboratory classes except computer based classes and experiments where no loose particles could be airborne.

16. Spillage of fuel, lubricating oil, coolant, water etc., should be cleaned up immediately. All guards must be in place before an engine or machine is started. Guards are NOT to be removed.

17. Wear heat-resistant gloves when using ovens and furnaces. Do not touch equipment or objects in hot areas without first checking that they are cool. Hot items should be clearly marked and kept in safe areas.

If you are doubtful about the safety of anything you are asked to do, you have a DUTY to refuse to do it until you have been properly instructed. If you come across a situation you think is unsafe, you have a DUTY to report it to a member of staff

Appendix 3

MSc Project (EMS715)
Coursework Assessment Structure & Mark Sheets
2024-2025

	Component of Assessment	Weighting %	UK Engineering Council Professional Standards for Engineers Accreditation of Higher Education Programmes (AHEP4) Learning Outcome Codes *Refer also to coursework assessment criteria.
1	**Directed Learning Assignment 1 (DLA1 – critical review)**	10%	M1, M2 & M4:
2	**Directed Learning Assignment 2 (DLA2 - Ethics)**	10%	M7
3	**Project Rationale, Aims and Objectives (RAO)**	10%	M2 & 17
4	**State of the Art report (SOTA)**	15%	M2, M4 & 17
5	**Final Report**	40%	M2, M3, M4, M5 & 17
6	**Viva Examination**	15%	M17
	Total	**100%**	

CW1 Mark Sheet DENM100: Directed Learning Assignment 1 (DLA1). CW1 Critical review of literature and application.

Criteria	Assessment Criteria CW1	Marks Range	Mark Awarded
1	**ALL THREE** completed DLA1 templates submitted with a coversheet in a single pdf format.	0-10	
2	**Part A** – accurately completed with article relevance clearly justified	0-15	
4	**Part B** – relevant and concise details extracted and summarised.	0-30	
5	**Part C** – **TWO** citations clearly identified with a clear and concise explanation of contribution to the article/literature.	0-15	
6	**Part D** – a clear, concise with logical justification and coherent argument – insightful presentation of ideas and argument beyond simply reporting information.	0-30	
		Total	

CW1: *Directed Learning Assessment 1 (DLA1 - Critical review of literature and application).*
AHEP4 M1, M2 & M4: *Explore and analyse engineering/scientific papers through a literature search i.e., a knowledge acquisition through a critical literature review. Understand, extract and apply key scientific/engineering principles and methodologies relevant to the project area of interest.*

CW2 Mark Sheet DENM100: Directed Learning Assignment 1 (DLA2). Ethics: Ethical Principles and Academic Integrity MSc (DLA2)

Criteria	Assessment Criteria CW2	Marks Range	Mark Awarded
1	Statement of Ethical Principles (Engineering Council)	0-100	**Online Assessment (QM+ Quiz)**
2	Ethics in the Engineering Profession (RAEng)		
3	Academic Integrity		
4	Ethics and Academic Misconduct		
		Total	

CW2*: Directed Learning Assessment 2 (DLA2: Ethics in Science & Engineering)*
AHEP4 M7*: Develop and awareness and understanding environmental and societal impact of scientific and engineering decision-making with professional ethical frameworks e.g., RAEng, Engineering Council.*

CW3 Mark Sheet DENM100:
MSc Rationale, Aims and Objectives A1 Size e-Poster.

Criteria	Assessment Criteria CW3	Marks Range	Mark Awarded
1	Clarity & Visual Impact of the Poster Design Logical layout and flow of text, images & figures.	0-10	
2	Imaginative and Creative Poster Design Use of a variety of visual and presentational techniques.	0-10	
4	Poster Content - RAO Rationale, aims and objectives are clear, relevant, focused and meaningful.	0-10	
5	Poster Content - Data, Figures, Illustrations and Diagrams. The use of data, figures and diagrams are relevant and purposeful.	0-10	
6	Outcomes - Findings & Implications Findings and implications of (actual or anticipated) findings and outcomes are clearly articulated.	0-10	
7	Purpose & Relevance (and their impact) Purpose and relevance of the project is clear, accessible and inclusive to both technical and non-technical audiences.	0-15	
8	Potential Benefits (and their impact) The benefits of the project and implications of project outcomes (actual or anticipated) are explained and illustrated in a clear and concise manner.	0-15	
9	Referencing Appropriate referencing is used in the poster and a list of key references are included in the appropriate format.	0-10	
		Total	

CW3*: Rationale, Aims and Objectives Poster*
AHEP4 M2: *Formulate, analyse and evaluate complex scientific/engineering principles to inform judgements on formulating a project rationale, aims and objectives (RAO) to extend research and generate new knowledge or outcomes.*
AHEP4 M16: *Function effectively as an individual and as a member of team working with academics, project supervisors and laboratory/workshop technical staff. Communicate the rationale, aims and objectives for an engineering/scientific project in an effective and concise manner in an A1 size project poster format for both a technical and non-technical audience.*
AHEP4 M17: *Effective use of the rationale, aims and objectives (RAO) and A1 poster to communicate complex engineering matters with technical and non-technical audiences, evaluating the effectiveness of the methods used*

CW4 Mark Sheet DENM100:
MSc Project State of the Art (SOTA).

Criteria		Assessment Criteria CW4	Marks Range	Mark Awarded
1	**Report Structure**	Well-structured and coherently written report.	0-7.5	
2		Consistent formatting of text, figures, tables, referencing and references.	0-7.5	
4		Sections are constructed in a logical and purposeful way with clear signposting for the reader	0-15	
5	**Technical Content**	Comprehensive, clear and concise review of the relevant literature pertinent to the project RAO.	0-10	
6		Awareness of developing technologies in field and implications for the extended research project.	0-10	
7		Critical engagement in relevant engineering & scientific concepts.	0-15	
8		Logical and justified presentation of arguments with implications for the extended research project clearly identified.	0-15	
9		Clear and accurate information provided.	0-10	
10		Appropriate use of figures, tables, data and references.	0-10	
			Total	

CW4: *Project State of the Art (SOTA)*
AHEP4 M2: *Formulate, analyse and evaluate complex scientific/engineering principles to inform judgements to support the project rationale, aims and objectives. Extend knowledge and understanding through extended literature review.*
AHEP4 M4: *Select and evaluate technical literature and other sources of information relevant to the project rationale, aims and objectives (RAO) to address complex engineering/scientific problems through an extended research project.*
AHEP4 M16: *Function effectively as an individual and as a member of team working with academics, project supervisors and laboratory/workshop technical staff. Use the extended engineering/scientific literature review to develop and communicate the rationale, aims and objectives (RAO) with project supervisors and academics.*
AHEP4 M17: *Generate a SOTA Report to help communicate effectively on complex engineering matters to technical and non-technical audiences to support and strengthen the project rationale, aims and objectives (RAO).*

CW5 Mark Sheet DENM100:
MSc Project Final Report (Extended Research Project).

Criteria		Assessment Criteria CW5	Marks Range	Mark Awarded
1	**BACKGROUND, INTRODUCTION and LITERATURE REVIEW - Quality of coverage and context (__/20)**	[Clear Fail]: Essentially no background material presented or few www references.	0-7	
		[Borderline Fail]: Scientific background poorly conveyed; absent key references/concepts.	8-9	
		[Pass]: Most scientific aims identified; some confusion relating to prioritisation and relevance.	10-11	
		[Merit]: Coherent narrative; almost all aims outlined.	12-13	
		[Exceptional]: Clear display of subject mastery; precise, coherent, structured; all key refs. cited	17-20	
2	**UNDERSTANDING & ANALYSIS - Scientific awareness, justification (__/30)**	[Clear Fail]: Demonstrates essentially no understanding of project; no meaningful analysis.	0-11	
		[Borderline Fail]: Very limited grasp of key scientific issues; little evidence of critical analysis.	12-14	
		[Pass]: Some comprehension of scientific issues; attempts to apply reason-based analysis.	15-17	
		[Merit]: Appreciation of scientific challenges; reasoned justification of strategic decisions taken	18-20	
		[Distinction]: Clear awareness of scientific challenges; logical approach to problem solving	21-25	
		[Exceptional]: Lucid presentation of nub of challenges faced; evidence-based reasoning and analysis.	26-30	
3	**QUALITY OF RESULTS & DISCUSSION - Extent and accuracy (__/30)**	[Clear Fail]: Quality of data doubtful; format unconventional	0-11	
		[Borderline Fail]: Presentation of data imprecise or inaccurate	12-14	
		[Pass]: Some useful standard data presented; format and presentation non-ideal	15-17	
		[Merit]: Useful data presented (or papers discussed); most	18-20	

		important details documented; some inconsistencies		
		[Distinction]: High quality results and discussion of publication standard (data from multiple papers compared); format largely adheres to convention	21-25	
		[Exceptional]: Outstanding high-quality results and discussion; substantial project progress mad	26-30	
4	**PRESENTATION - Structure, clarity, written style, quality of English (__/20)**	[Clear Fail]: Lacking any apparent order; English usage very poor; no links between text and figures/tables	0-7	
		[Borderline Fail]: Minimal organisation of material; difficult to follow; unclear; poor English	8-9	
		[Pass]: Apparent structure into sections; English acceptable	10-11	
		[Merit]: English good; organisation of material into logical sections; clear style	12-13	
		[Distinction]: Essentially error free; clear logical construction; balanced presentation of message	14-16	
		[Exceptional]: Flawless English; clear, logical structure; engaging style; well-developed message	17-20	
			Total	

CW5: *Project Final Report (Extended Research Project)*
AHEP4 M2: *Generate a final project report to build on complex scientific/engineering principles applied to the research project to show how an analysis of project outcomes support project conclusions.*
AHEP4 M3: *Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.*
AHEP4 M4: *Update and develop technical literature reviews and other sources of information to inform project discussions relating to complex engineering/scientific problems beyond the limitations and constraints of the final project report.*
AHEP4 M5: *Provide a final report that contains design solutions for complex problems that meet a combination of societal, user, business and customer needs, as appropriate.*
AHEP4 M16*: Function effectively as an individual and as a member of team working with academics, project supervisors and laboratory/workshop technical staff. Use the extended engineering/scientific literature review to develop and communicate the rationale, aims and objectives (RAO) to project supervisors and academics.*
AHEP4 M17: *Generate a Final Report to help communicate effectively on complex engineering matters to technical and non-technical audiences to show fulfilment of project rationale, aims and objectives (RAO).*

CW6 Mark Sheet DENM100:
MSc Project Viva (Oral Examination).

Criteria		Assessment Criteria CW6	Marks Range	Mark Awarded
1	**SUMMARY and PRESENTING the PROJECT __/50**	<25 [Fail]	0-24	
		25-29 [Pass]	25-29	
		30-34 [Merit]	30-34	
		35-40 [Distinction]	35-40	
		41-50 [Exceptional]	41-50	
2	**VIVA DISCUSSION - Extent, depth and accuracy __/50**	<25 [Fail]	0-24	
		25-29 [Pass]	25-29	
		30-34 [Merit]	30-34	
		35-40 [Distinction]	35-40	
		41-50 [Exceptional]	41-50	
			Total	

CW6: *Oral examination – Viva*
AHEP4 M16*: Participate in an oral presentation and review of the project effectively as an individual, and as a member of the project supervision team.*
AHEP4 M17: *Under Viva conditions, communicate effectively on complex engineering matters with technical and non-technical audiences. Explain and justify project via a presentation and through responding respond to questions and queries that challenge the project rationale, approach, methodology, results/outcomes and conclusions.*

1. Viva will be held with both an **examiner (marker 2) and your supervisor (marker 1**).
2. The examiner will lead the viva, but both your examiner and supervisor will separately assess your viva (oral examination).
3. It is recommended that you **prepare a concise 2-3 minute presentation** (**but 3 minutes <u>maximum</u>**) to summarise your extended research project (e.g., rationale, aim, objectives, specific problem(s) or context, method/approach, justification, outcomes (results, findings and outcomes). A concise summary of your extended research purpose and outcomes will help you prepare for your viva and it will also give you an opportunity to cover the salient points from from your project report for your examiner at the start of your oral examination (viva). PowerPoint can be used, but you can use other formats for your project summary. **Remember to upload a pdf file of your viva presentation to <u>both</u> viva submission points before your viva.**
4. Remember that your project summary needs to be short, clear and concise e.g., what you did; how and why you did it; what was the outcome (results & conclusion); why was it useful and how did you address/meet the project RAO.
5. The **oral examination (viva) will take 15-20 minutes**, and during this time you will be asked questions about your project and the final report. Your examiner and supervisor **will already have received your final report so it is recommended that also you have your report to hand during the viva** so that, if necessary, you can refer to sections of the report to help support and justify your answers, explanations and reasoning.
6. The viva is an **opportunity to show the work you did and demonstrate your knowledge, understanding, analysis** driven by clear purpose and rationale.
7. The **viva assessment criteria** will be based on the following areas (see below).

Appendix 4

UK Engineering Council Professional Standards for Engineers
Accreditation of Higher Education Programmes (AHEP4) Learning Outcome Code Descriptots

M-Level Code Descriptors

M1 Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study.

M2 Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles.

M3 Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.

M4 Select and evaluate technical literature and other sources of information to address complex problems.

M5 Design solutions for complex problems that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards.

M7 Evaluate the environmental and societal impact of solutions to complex problems and minimise adverse impacts.

M17 Communicate effectively on complex engineering matters with technical and non-technical audiences.